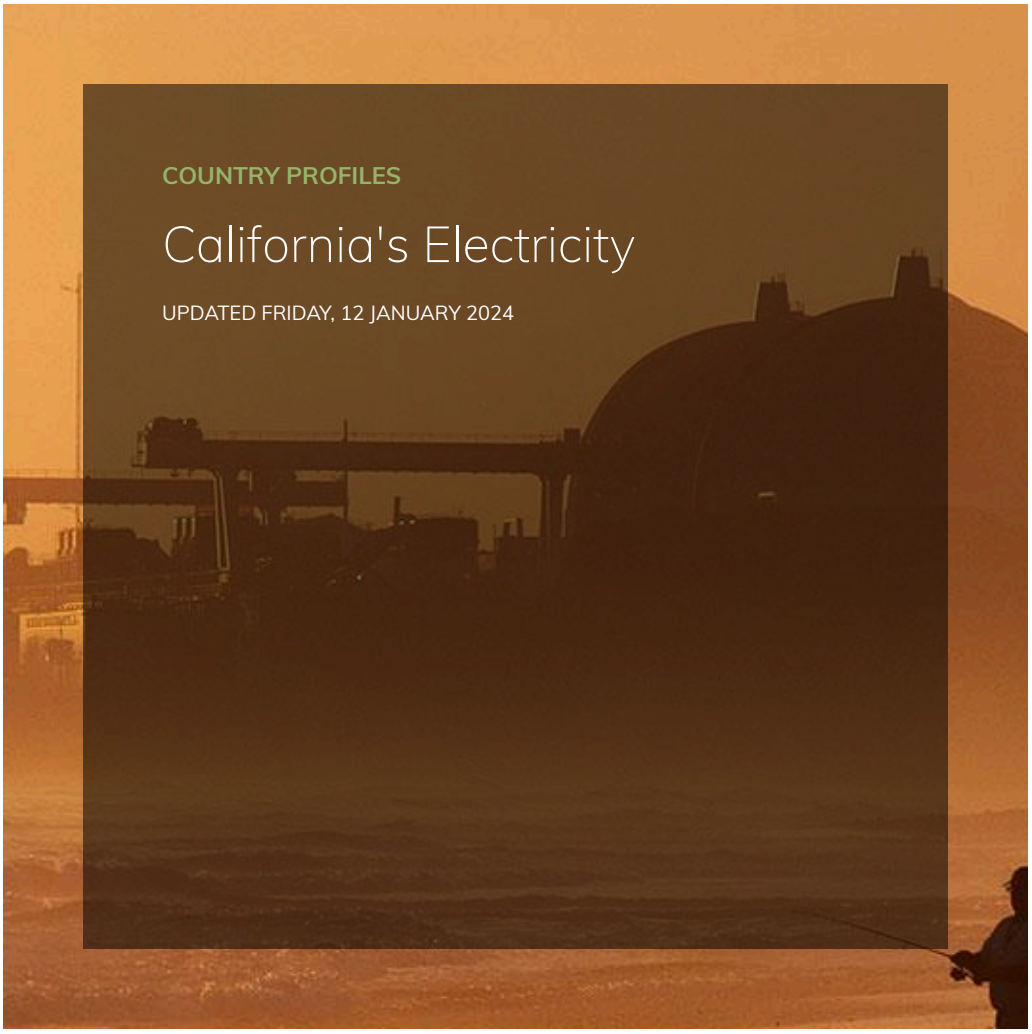


- California has a large and growing population and a flourishing economy, with a major high-technology sector.
- Electricity demand has been rising steadily for many years, but it lacks sufficient reliable in-state supply.
- In 2000-01 there was an acute electricity supply crisis, triggered by several factors.
- In the summer of 2020 there was another supply crisis, with an emergency being declared by the system

Privacy - Terms

operator.

- California has set the target of generating all the state's electricity from low-carbon sources by 2045.

California has a population of some 39 million, which grew by over 25% through the 1980s and 12% through the 1990s. It is expected to reach 55 million by 2050. Its economy is the world's fifth largest, and includes a major high-technology sector. It produces 13% of US gross domestic product (GDP).

In August 2018 the state assembly voted to produce 60% of electricity from renewables by 2030 and produce all the state's electricity without fossil fuels by 2045. At present, just under half is supplied by natural gas, one-tenth from nuclear power and up to one-fifth from in-state hydro (depending on rainfall). The state has set a goal of reducing its carbon dioxide emissions by 80% of 1990 levels by 2050, as required by its Global Warming Solutions Act of 2006 and a 2005 executive order.

California Energy Commission (CEC) data for 2021 showed in-state generation of 194 TWh from 82.9 GWe of installed capacity and net imports of 83.6 TWh to give a total consumption of 278 TWh. In-state generation comprised: 16.5 TWh (8%) nuclear; 97.4 TWh (50%) natural gas; 33.3 TWh (17%) solar; 15.2 TWh (8%) wind; 12.0 TWh (6%) large hydro; 11.1 TWh (6%) geothermal; 5.4 TWh (3%) biomass; 2.5 TWh (1%) small hydro; and 0.3 TWh coal. The imports were 32.6 TWh from Pacific Northwest and 51.1 TWh from Southwest, the latter including 15.7 TWh from coal and gas. About 12 GWe of gas-fired capacity was retired in the eight years to mid-2020.

In 2021, around 30% of California's electricity supply came from outside the state. Electricity supplied from out-of-state coal-fired power plants has decreased due to the state's Global Warming Solutions Act of 2006 that requires Californian utilities to limit new long-term financial investments in base-load generation to only those power plants that meet California emissions performance standards.

The state played a significant role in the early development of US nuclear power. Until mid-2013 it had two nuclear power plants: Diablo Canyon and San Onofre – with four PWR reactors totalling 4390 MWe capacity. These are designed to survive large earthquakes. However, San Onofre has now closed down, taking 2150 MWe offline (see section on The 2013 San Onofre saga below). The continued operation of Diablo Canyon has been in

question, and in June 2016 closure of the two reactors in 2024 & 2025 was announced. However in September 2022 the California Assembly passed Senate Bill 846, enabling the Diablo Canyon nuclear power plant to remain operational for up to five years longer than originally planned (see section on Diablo Canyon countdown below).

Aside from some large wind farms, hardly any generating capacity was built in California in the 20 years to 2000. Development of almost all new capacity was prevented by environmental activism, despite annual demand growing at a rate of about 2% per year. By 2001, some 80% of California's generating plants were older than 35 years (the two largest gas-fired plants were 45 years old). Some 3000 MWe of gas combined cycle plant came online by the end of 2001 and a further 8400 MWe from then to the end of 2005. From 2001 to 2015, in-state capacity increased from 53.3 GWe to 79.4 GWe, including a lot of intermittent renewables, though in-state generation declined marginally.

From 2002 to 2006 in-state generation from coal dropped by 36% due to the closure in 2005 of the Mohave plant (out-of-state but Californian-owned, hence deemed 'in-state'), and that from natural gas rose by 18% in line with a 26% increase in installed capacity (after dropping back to pre-2000 levels in 2002). Power prices rose much more in California than the rest of USA from 2011 to 2019 and are now about 60% above the US average.

In mid-2016 the California Independent System Operator (CAISO) said in a 700-page report that expanding its operations to include more Western states would result in a more efficient electricity grid, reduce greenhouse gas emissions across the west, as well as meet or exceed the state’s goal of obtaining half of its energy from renewable sources. The study showed that an 11-state regional market would cut costs by allowing generators to sell excess electricity more easily across state lines, as well as letting California import larger amounts of renewable energy from neighbouring states. CAISO said that California was set to have a surplus of 13 GWe of renewable energy by 2025 that would have to be curtailed when peak generation exceeded demand. Expanding the ISO territory would enable this to be shared or dumped interstate without shutting down turbines. The last public meeting was held in October 2016; however, as of early 2023, there has been no legislation requiring a regional grid.

In 2019 the California Public Utilities Commission (CPUC) authorized the three utilities to procure around 3.3 GWe of new capacity to deal with potential shortfalls, and extend the operating

lifetimes of 4.8 GWe of natural gas plants. The new capacity was scheduled to begin coming online in 2021.

In mid-2021, the CPUC ordered utilities to procure 11.5 GWe of new electricity capacity from preferred resources (including renewables and zero-emiitting sources, as well as energy efficiency and demand response measures), to come online between 2023 and 2026. The CPUC in February 2023 ordered the procurement of a further 4 GWe.

A target of around 25.5 GWe of new renewables plus around 15 GWe of new storage and demand-response resources by 2032 was set by the CPUC in February 2022.

The 2000-01 crisis

Compounding the long-term problem, towards the end of 2000 the state had a lot of its generating capacity off-line, mostly catching up with maintenance deferred from peak summer load conditions. California thus faced severe power constraints and these continued through the winter into 2001.

Several plants, totalling 2700 MWe, had used up their annual pollution credits so could not restart without severe fines. In particular, three gas-fired plants (2000 MWe) were shut down after the south coast Air Quality Management District required them to install emission control equipment for NOx (oxides of nitrogen). As the crisis developed, CAISO called them back into service, but they were required to obtain NOx emission credits to cover the short-term impact of this. The price of such credits soared.

Furthermore, a dry summer had reduced hydroelectric availability in the US Northwest. Interstate coal and nuclear plants helped, and local gas-fired plants met some of the shortfall, but demand forced gas prices to double. In 1999, gas provided 31% of the state's electricity, including imports (37% in 2000 and 40% in 2001).

As a result, wholesale electricity prices throughout the US West soared to unprecedented levels — briefly reaching $750/MWh (75 ¢/kWh). In December 2000, it averaged $377/MWh and the weekly average ranged from $198 to $350/MWh in January 2001. (Prices in 1998-99 had been $24-28/MWh.) Electric utilities generally experienced a quadrupling of wholesale prices from generators but they had their own prices capped at 16 cents/kWh and consequently suffered about $12 billion in losses over just six

months. The state government stepped in to bail out the two largest utilities and re-regulate the system.

Three west coast aluminium smelters with long-term power supply contracts closed until 2002, and made more money selling their electricity entitlements than they could from aluminium, while employees were sent home on full pay. In one case, power bought for 2.25 ¢/kWh was being sold for 55 ¢/kWh.

The supply crisis, with rolling blackouts, ran from mid-2000 to late February 2001. The shortfall in generating capacity was widely seen as being due to years of weak government appeasing extreme environmentalism. It forced renewed interest in building substantial coal and nuclear capacity to meet base-load demand. Support for building new nuclear power plants in the USA generally jumped as a result of the Californian crisis. In the west of the country, opinion that "we should definitely build more nuclear energy plants in the future" rose from 33% to 52%, nationwide it rose from 42% to 51%, compared with October 1999. More than two-thirds said that nuclear energy should play an important role in meeting future US energy needs.

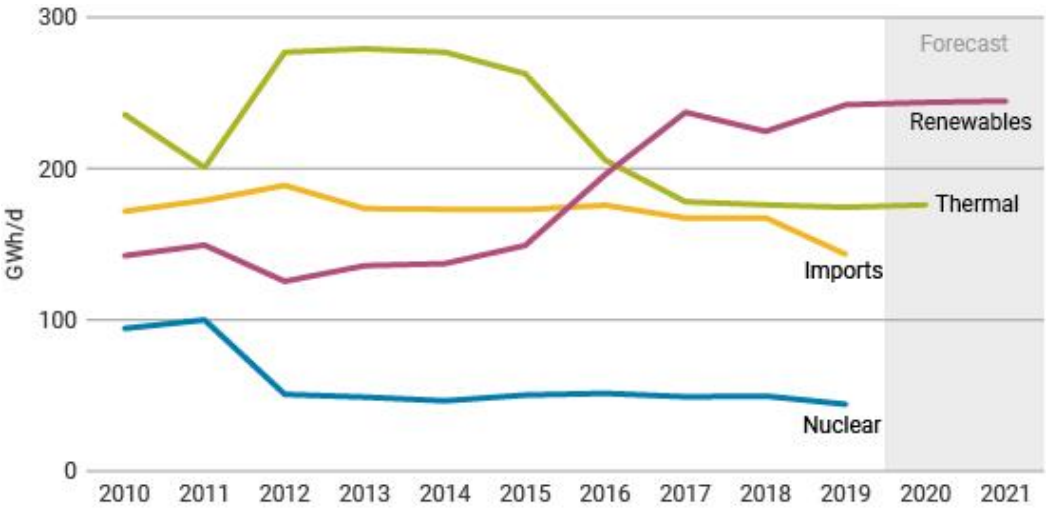
In the event, demand reductions and load shifting through mid 2001 meant that summer blackouts were averted. Gas-fired capacity was much more fully utilised than before or after 2000-01.

The 2020 crisis

In August 2020 California again experienced power shortages with rolling blackouts. Due to a severe, but not extraordinary, heatwave coinciding with little wind, day-ahead electricity prices spiked at above $1000/MWh. Demand climbed to 49 GWe. CAISO declared a high-level emergency for the first time in 20 years and ordered consumers to reduce electricity demand to keep the power on as much as possible. Imports of power were constrained due to high demand interstate.

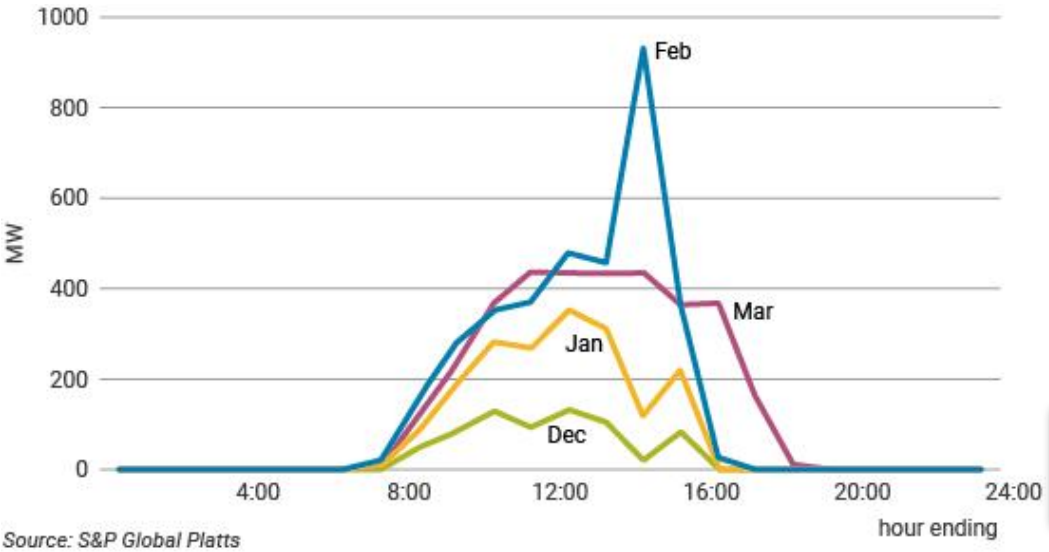
Governor Gavin Newsome said that California's transition away from fossil fuels (notably gas) was a contributing factor to the state's rolling blackouts. The displacement of fossil fuel by solar power and other forms of green energy as "a moral and ethical imperative" had led to what he called "gaps" in the energy grid's reliability. "Collectively, energy regulators failed to anticipate this event and to take necessary actions to ensure reliable power to Californians," he said in a letter to CAISO. CAISO attributed the blackouts to the unexpected loss of a 470 MWe gas-fired power

plant, as well as a shortfall of nearly 1000 MWe expected from wind power. The head of CAISO said that the non-profit corporation had repeatedly warned of a reliability gap, and that the state would have to increase dispatchable capacity. Thus the root cause of the emergency was the state's politically-determined overreliance on solar and wind power.



Source: Cal-ISO, S&P Global Market Intelligence

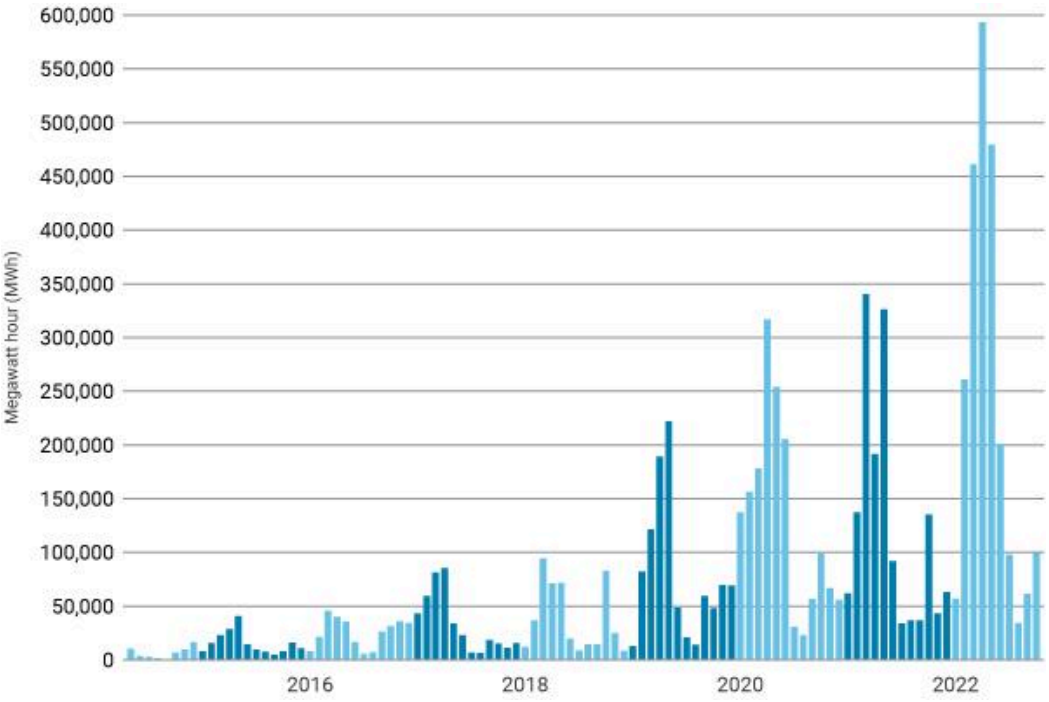
Along with acute shortages of supply when needed, electricity curtailments have been rising every year, driven by growth in solar power to meet the state's aggressive clean energy goals. They total up to about 3% of production, and are mainly solar. CASIO curtailed 223,000 MWh of wind and solar in May 2019, and then curtailed 138,000 MWh in January 2020 and 157,000 MWh in February, before the coronavirus pandemic reduced demand. In March 300,000 MWh was curtailed. On 1 April 2020, 17,000 MWe was reported to be curtailed, reducing the spot price to $6/MWh. In March 2021 curtailment reached 342,000 MWh and peaked in April 2022 at 596,000 MWh. Some curtailment is due to oversupply, and some – localized – due to grid congestion. Managing this accounts for the majority of ISO curtailments.



Source: S&P Global Platts

The power crisis coincided with major wildfires in August 2020 which caused the governor also to declare a state of emergency on that account.

California wind and solar curtailments



Deregulation

Much newspaper coverage of the earlier Californian power crisis has pointed to deregulation as a factor, if not a cause.

Before 'deregulation', electric utilities, which have a legal obligation to serve their customers' demands, could build plants regardless of the expense and recover costs from customers. In 1996, utilities owned 81% of the total generating capacity and the average retail price was 9.5 cents/kWh, the tenth highest in the USA. This arrangement locked in certain inefficiencies, and when deregulation loomed it raised the question of how utilities would recover their 'stranded costs', mostly the capital component which could not be amortised with expected lower electricity prices. Elaborate mechanisms were put into place to cover these, but there were conditions imposed to ensure that utilities did not exploit the situation.

Under the Electric Utility Industry Restructuring Act in 1996 the Californian government put into place a deregulation scheme which sought to bring competition into generation, attracting needed investment, while leaving transmission and distribution as regulated monopolies. This required the major utilities to divest at

least half their major generation assets, so that their ownership fell to 46% of the total capacity.

The scheme also prevented them from entering long-term hedging contracts that would limit the risk of large price movements, forced them to buy electricity at market rates from a centralised pool, and on top of all this committed the two main utilities to retailing the electricity at fixed 1996 prices until March 2002 regardless of the cost of wholesale purchase. The price cap provision incorporates a transition charge which is the mechanism for utilities to recover stranded costs.

Thus there was not so much deregulation as a much less effective form of regulation. The need for long-term contracts enabling generators to build and maintain adequate capacity was emphasised, as was the need for adequate reserves which consumers had to be prepared to pay for maintaining.

The 2013 San Onofre saga

Early in 2013 California was focused on avoiding an electricity crisis partly arising from its growing reliance on wind and solar power, and partly from one nuclear power station being shut down with steam generator problems.

Units 2&3 of Southern California Edison's San Onofre Nuclear Generating Station (SONGS) had been shut down since January 2012 because of faults in their new steam generators supplied by Mitsubishi Heavy Industries and installed over 2009-11. A Nuclear Regulatory Commission investigation team pointed to "faulty computer modelling" and "manufacturing issues" as contributing to the rapid deterioration of the steam generator tubing. In total some 386 tubes had thinned by more than 35% from their original state – a level that required mandatory plugging – while hundreds more were plugged as a precaution. Each steam generator contains 9727 tubes and is designed to cope with the loss of some of these throughout its 30-40 year life, however the speed of degradation since 2010 was surprising and was said to be a possible safety issue if a large failure were to suddenly occur. Both units remained shut down with no clear timetable for their return to service, taking 2150 MWe net offline. Despite three independent engineering reviews confirming the steam generators' safety at 70% capacity, the NRC delayed giving approval for this for some eight months. Hence in June 2013 SC Edison decided to retire them permanently due to the regulatory delay and uncertainty in bringing

the 40-year old units back into service. Fuel was removed by mid-July 2013.

SC Edison and Mitsubishi were at odds over terms of the steam generator warranty. Edison filed warranty claims of $139 million – just over the warranty's stated limit – while asserting that liability limitations in the contract did not apply. Long-term solutions to rapid degradation of the plant's steam generators were being pursued independently of a proposal to restart the plant at partial power. The company indicated that without restart approval, it might be compelled to decommission the plant and write it off, which it then implemented. In July 2013 Southern California Edison served a formal Notice of Dispute on Mitsubishi Heavy Industries, Ltd., and Mitsubishi Nuclear Energy Systems, which seeks to hold Mitsubishi accountable for designing and manufacturing defective replacement steam generators which were warranted for 20 years, and claiming over $4 billion through mediation by the International Chamber of Commerce. In July 2015 SC Edison said that it was increasing its claim against MHI to $7.57 billion. MHI responded saying that "The allegations and demands made by [SCE] disregard the history of the contract negotiations and performance and are factually incorrect, legally unsound, and inappropriate."

In October 2015 the owners of SONGS reached a $400 million settlement with Nuclear Electric Insurance Limited for outages caused by the failures of the replacement steam generators. SCE said it continued to pursue arbitration claims against MHI and Mitsubishi Nuclear Energy Systems for failure of the steam generators. In March 2017 the International Chamber of Commerce awarded $125 million – the warranty liability limit in the original contract – to SCE and the 22% minority owners, less $58 million legal fees and costs to be paid to MHI. SCE ended up with $52 million from the warranty claim.

The plant is midway between Los Angeles and San Diego, and played an important role in grid stability for the ISO. With both San Onofre units off line since January 2012, California's wholesale electricity prices suffered. The US Energy Information Administration (EIA) reported a 59% increase in wholesale power prices in the state over the first half of 2013, which it ascribed largely to the extended outages at the two units. The situation also caused a "large and unusual" separation in power prices between the northern and southern Californian electricity grids, which have historically tracked each other closely.

The 2013 Integrated Energy Policy Report from the California Energy Commission said that the state must find replacement low-carbon capacity for San Onofre to offset the emissions avoided by that plant and address the effects of its closure on the reliability of the state's electric grid. In particular the demand response goals set in the early 2000s needed to be tackled. To compensate for the closure of SONGS and maintain grid reliability, in October 2013 the California Public Utilities Commission established an energy storage target of 1325 MWe for Pacific Gas and Electric Company, Southern California Edison, and San Diego Gas & Electric by 2020, with installations required no later than the end of 2024. As of April 2023 California had just over 4500 MWe of installed battery storage capacity.

Diablo Canyon countdown

Pacific Gas & Electricity's Diablo Canyon nuclear plant continued in operation through most of the drama with SONGS. However, in June 2016 PG&E announced that its two reactors, of 1138 and 1118 MWe net would close down in 2024 and 2025 after only 40 years' service, rather than continuing with a licence renewal application which would take them to 2045 – a 20-year extension like most other US nuclear plants. The new proposal was subject to confirmation that the company's $2 billion investment in Diablo Canyon will be recovered by the time the plant closes and that the costs of replacement greenhouse gas-free capacity can be passed onto consumers (it is a regulated plant, not subject to wholesale price competition from gas).

PG&E said that the decision was agreed to by a coalition of labour and environmental groups, including Friends of the Earth and the Natural Resources Defense Council. Implementation of the shutdown proposal was also contingent on the State Lands Commission approving a lease extension beyond 2018 for Diablo Canyon's cooling water intake and discharge facilities, which was promptly given, and the CPUC's approval of the proposal to replace the plant with renewable energy resources.

The parties to the 2016 joint proposal (to shut down the plant in 2025) said: "This proposal recognizes the value of GHG-free nuclear power as an important bridge strategy to help ensure that power remains affordable and reliable" but did not say why continuing operation would not achieve this. They acknowledged "the challenge of managing overgeneration and intermittency conditions under a resource portfolio increasingly

influenced by solar and wind production" but did not indicate how less nuclear and more solar and wind would help address this. They said the proposal would "impact the efficient and reliable balancing of load and resources in PG&E's service territory," which then already faced stability challenges in integrating intermittent renewables. They added: "The retirement of Diablo Canyon may have impacts on system ramping and local reliability, and must be resolved by the CPUC through its IRP process [integrated resource planning process for regulated load-serving entities], in collaboration with the CAISO [California Independent System Operator]."

The main driver deterring PG&E from seeking a 20-year operating licence extension prior to the 2016 joint proposal was the 2015 renewable portfolio standard (RPS) of producing 50% of its electricity from qualified renewable energy sources by 2030. PG&E's model for the future cost of operating Diablo Canyon indicated that the cost per kilowatt hour was going to almost double, since the company would be forced to lower the amount of power it could produce from the plant in order to meet the state's requirement. Dropping the capacity factor from 92% to, say, 50% would have virtually doubled the price per kilowatt hour since costs were largely fixed.

Coupled with consideration of the RPS, PG&E's decision was informed by the presumed need for some refurbishment to take the reactors to 2045 if licences were renewed. One aspect of this was a threat that very expensive cooling towers would be needed due to State Water Resources Control Board requirements, rather than continuing with direct once-through cooling from the ocean.

Despite these negative indicators, there was a significant campaign to keep the plant open since the announcement of the 2016 joint proposal. Late in 2018 Californians for Green Nuclear Power (CGNP) was an adverse intervenor before the California Public Utilities Commission (CPUC) relating to PG&E's requesting permission from the CPUC to voluntarily close "the safe and highly-performing Diablo Canyon Power Plant" in 2025. Then in October 2020 CGNP asked the Federal Energy Regulatory Commission (FERC) to launch an investigation into whether shutting down Diablo Canyon violated federal reliability standards and whether two regulatory bodies were negligent in ensuring reliability. The group contended that the 2020 blackouts were not an isolated incident but a consequence of "much larger systemic reliability challenges that will only be made worse by the voluntary closure of [Diablo Canyon]."

In November 2021, a report from Stanford University, the Massachusetts Institute of Technology and LucidCatalyst outlined the value of delaying the retirement of the Diablo Canyon plant. The report concluded that delaying the retirement of the plant by ten years would “significantly reduce California’s use of natural gas for electricity production from 2025 to 2035 by approximately 10.2 TWh per year,” reducing California’s power sector carbon emissions by more than 10% from 2017 levels. The study also mentioned the plant's potential for powering water desalination and hydrogen fuel production. In the same month, the American Nuclear Society (ANS) warned that the premature shutdown of units 1&2 would "inflict grave harm to California's economy and environment." The ANS warned that premature closure of the plant would deprive California of its largest carbon-free energy source, whilst weakening its already fragile power grid.

In February 2022 the San Luis Obispo County Board agreed to write to state Governor Gavin Newsom to urge him to work with PG&E to ensure it had access to all the necessary permits to keep the plant operational. Three members of the Board of Governors voted in favour of the motion, with one voting against, and former PG&E employee Dawn Ortiz-Legg recusing. In May 2022 a study commissioned by Carbon Free California found that there was strong state-wide support for the continued use of the Diablo Canyon plant. It found that 58% of voters across the state supported continued operation of the plant, with 32% opposing it. There was even more support in the area surrounding the plant, with 74% of participants in San Luis Obispo County voting in favour of continued operation of Diablo Canyon, with 56% of respondents "strongly" supporting it.

In August 2022 California Governor Gavin Newsom proposed to keep the Diablo Canyon nuclear plant operational for an additional five-to-ten years. The draft legislation would extend up to $1.4 billion in loans to PG&E to keep the two units online beyond their planned 2024 and 2025 closure dates and into the 2030s.

In September 2022 the California Assembly passed Senate Bill 846, enabling the Diablo Canyon plant to remain operational for up to five years longer than originally planned. The Assembly passed the bill by 69 votes, before returning back to the Senate for the final vote where senators voted 31-1 in favor of the bill. The utility still had to receive an operating licence extension from the Nuclear Regulatory Commission (NRC) to operate beyond the existing operating licence expiry dates.

In January 2023 the US Nuclear Regulatory Commission rejected PG&E's request to resume its review of a previously withdrawn licence renewal application. Subsequently PG&E planned to submit a new licence renewal application.

In March 2023 the NRC announced the approval of PG&E's exemption request, meaning that units 1&2 would be allowed to operate past 2024 and 2025 respectively. The licence renewal application for the two units was accepted by the NRC in December 2023.

Diablo Canyon (PG&E)

New nuclear capacity

A 1976 state law prohibits construction of new nuclear power plants in California until a means of disposal of high-level nuclear waste is approved. A bill to repeal this moratorium was voted down in April 2007.

In September 2007, the California Republican party voted unanimously to work to remove the prohibition on new nuclear

power plant construction. However, to date no progress has been made towards ending the moratorium on new nuclear build.

Earlier in 2007 a group sought the involvement of UniStar Nuclear (then a joint venture of Constellation Energy and Areva Inc, now wholly owned by EDF) to investigate building one or two of Areva's 1600 MWe EPR units at Fresno.

In July 2011 a study from the California Council on Science and Technology (CCST) called for an almost tenfold increase in the state's nuclear energy capacity by 2050. The report, California's Energy Future - Powering California with Nuclear Energy, included a 'Realistic Model' scenario that assumed that the state's energy demand in 2050 would be 510 TWh/yr. It also assumed that nuclear electricity would be used as the base-load power source, that nuclear plants would have a 90% capacity factor, and that nuclear power would provide two-thirds of the state's electricity, the rest coming from renewables as required by the Global Warming Solutions Act of 2006. The scenario would require about 44 GWe of nuclear capacity.

This nuclear energy report was produced as follow-up to a wider CCST report, California's Energy Future: The View to 2050 (May 2011). This found that: "Nuclear power provides reliable base-load power with very low emissions and can offset variability issues incurred by renewables, but faces obstacles with current public policy and public opinion. By law, new nuclear power in California is currently predicated on a solution for nuclear waste." However, "If electric generation is predominantly intermittent renewable power, using natural gas to firm the power would likely result in greenhouse gas emissions that would alone exceed the 2050 target for the entire economy." The CCST prepared its reports for the California Energy Commission.

Notes & references

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